

Figure 1: OSGi layered model

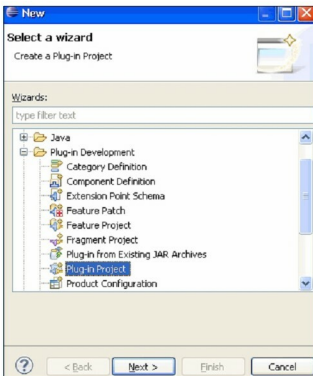


Figure 2: New project

Java programming language. The collection of underlying frameworks allows dynamic component re-usability, vast UI customisation, plugging and unplugging modules based on user requirements, and provides a rich client-side application.

Originally, the Eclipse platform was designed to serve as an open tool platform. Starting with Eclipse V3.0, it was re-architected so that its components could be used to build any client application. The minimal set of plug-ins needed to build a rich client application is collectively called the Rich Client Platform or RCP.

The framework has the ability to deploy native GUI applications to a variety of desktop operating systems such as Windows, Linux and Mac OSX, and an integrated update mechanism for deploying desktop applications from a central server. The RCP uses SWT and JFace from the Plug-In Development Environment (PDE). The Standard

Widget Toolkit (SWT) is a widget toolkit for Java developers with portable API for text controls, radio buttons, etc, used to create RCP user interfaces. There is tight integration with the underlying native OS GUI platform, which allows the toolkit to immediately reflect any changes in the underlying OS GUI look and feel. On the other hand, the JFace UI framework provides higher-level application constructs for supporting dialogues, wizards, actions, user preferences and widget management. The JFace APIs are independent of the workbench extension points and APIs. JFace is designed to provide common application UI functionality on top of the SWT library.

Such lower-level frameworks are integrated to form the workbench and structured to provide a customisable portal, where users can add the contents in the places defined.

What do I need to build an RCP application?

First, Java version 1.6 or above needs to be installed. To develop the plug-ins, you need the Eclipse platform. For Eclipse installation, the latest official build can be obtained from <http://download.eclipse.org/eclipse/downloads/>

Get set and go

Before we start with simple tool development, let's get familiar with two configuration files.

MANIFEST.MF: This contains the OSGi configuration information. The file describes Java packages that can be used by other plug-ins and their dependencies.

plugin.xml: This offers the possibility of defining extension points and extensions.

Let's now create a simple RCP application. In the Eclipse IDE, go to **File-> New Project**. From the list, select **'Plug-In Project'**.

Provide the project name as *com.lfy.rcp.first* and press **Next**. Make sure you chose **'Yes'** for **Rich Client Application** when prompted. Now press **Next** and select the template **'Hello RCP'**.

Press **Next**, select **'Add branding'** and then press **Finish**.

Take a look at the left pane. The view **'Package Explorer'** shows up a directory structure with source files, configuration files and libraries. Under source files, you will see some important files that are required for the RCP application development. You may also see **MANIFEST.MF** and **Plugin.xml** listed in the Package Explorer.

We are now going to start the RCP application. Let's execute our code in the IDE using a run configuration. A run configuration defines which application or product to launch, and which bundles are to be made available during runtime. In the **Overview** tab on the central pane, click on the **Launch an Eclipse application** link.

Figure 3 shows the output after running your first RCP application.

This isn't enough. We need to add contents in the form of views and menus. So let's continue to use the same RCP project. Select **'plugin.xml'** and click the tab **'Extensions'**. Add *org.eclipse.ui.views* and create a new view.

Provide an ID (say *com.lfy.rcp.first.MyView*) and name for the view. Further click on the class to create a Java class for the view. An editor opens up with the chosen class name. Furnish the